

SYNOPSIS

The objective of the “**AI-POWERED THREAT INTELLIGENCE IMPACT ANALYZER FOR ORGANIZATIONAL SERVICES**” is to provide an intelligent cybersecurity solution that automatically collects and analyzes threat intelligence from trusted cybersecurity sources. The system identifies newly published vulnerabilities, malware campaigns, Indicators of Compromise (IOCs), and cyber threats, compares them with an organization’s infrastructure, and determines whether its systems or services are at risk.

The front end of the project is a React.js-based web dashboard and the back end is developed using **Node.js with Express.js**, as the back end. **PostgreSQL** is used as the database for storing organizational assets. The operating environment used is **Windows 11**.

The existing system requires security analysts to manually monitor cybersecurity websites, vendor advisories, and vulnerability databases to determine whether newly published threats affect their organization. This process is time-consuming. In the proposed system, threat intelligence is automatically collected, analyzed using Artificial Intelligence, compared with organizational assets, and evaluated to identify affected services, prioritize risks, and recommend mitigation measures efficiently.

The project consists of six modules. **User Authentication Module** provides secure login and access control for authorized users. **Asset Inventory Management Module** stores organizational systems, software, and service information. **Threat Intelligence Collection Module** gathers cybersecurity reports from public threat intelligence sources. **AI Threat Analysis Module** uses Google Gemini to summarize reports, identify vulnerabilities, extract Indicators of Compromise (IOCs), and determine threat severity. **Impact Analysis Module** compares extracted threat information with organizational assets to identify affected services and evaluate business impact. **Reporting and Dashboard Module** presents threat summaries, impact analysis, risk levels, mitigation recommendations, and generates comprehensive security reports.